CMPE 258 • DEEP LEARNING • SHORT STORY

# One Model to Forecast Them All

*The Rise of Time-Series Foundation Models*

A review of **"Empowering Time Series Analysis with Foundation Models: A Comprehensive Survey"**

Ye, Yu, Zhang, Wang, Li & Tsung — arXiv:2405.02358v3 (2025)

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*Plus a hands-on reproduction of the survey's core claim — and why 2025's benchmarks can't be fully trusted.*

MOTIVATION

# Time series was the last modality to get a foundation model

| Text | Images | Audio | Time series |
|---|---|---|---|
| GPT, Llama | CLIP, SAM | Whisper | ??? until 2024 |

**Why time series resisted the “pre-train once, reuse everywhere” recipe:**

1. **Data is scarce & siloed** — No internet-scale firehose of signals. Useful series hide in private hospital, factory and bank silos.
2. **Every dataset has a different shape** — Different sampling rates, channel counts, and meanings — the survey calls it “semantic variability.”
3. **Classical methods were already strong** — ARIMA and exponential smoothing are fast and hard to beat on a single clean series.

THE CORE IDEA

# A foundation model: pre-train once, reuse everywhere

## What is a foundation model?

A large network pre-trained on a huge, broad dataset, then reused — often with no further training — across many tasks.

For time series the tasks are: forecasting, imputation (gap-filling), classification, and anomaly detection.

THE SURVEY

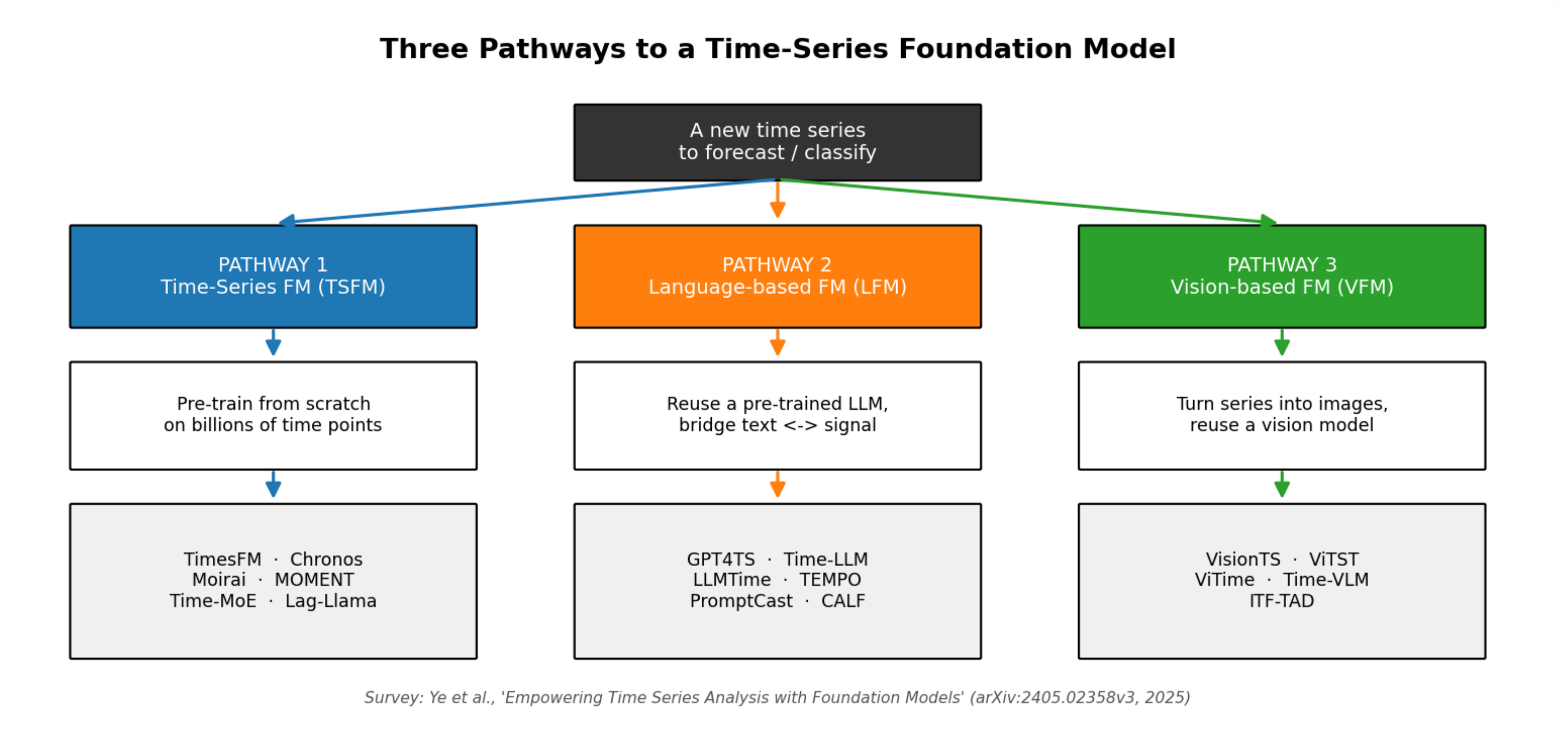
Ye et al. (2025) map 88 papers — the only survey covering all three build pathways under one “modality-aware, challenge-oriented” lens.

THE QUESTION THIS TALK ANSWERS

*Can a model pre-trained once forecast a series it has never seen — with zero training on it — and still beat models tuned to that series?*

This is the “zero-shot” promise. Slides 11–14 put it to the test on real data.

# Three pathways to a time-series foundation model




## 1 · TSFM

Pre-train from scratch on billions of real time points.

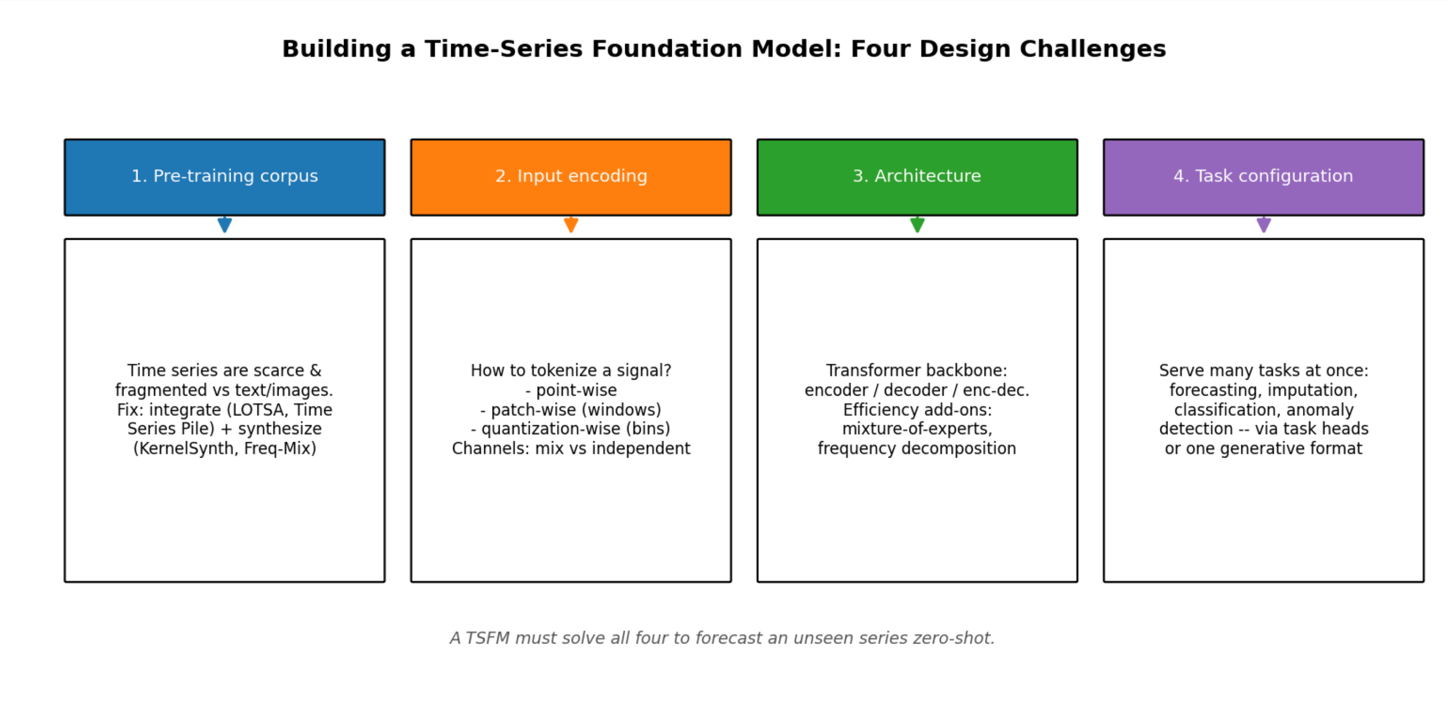
## 2 · LFM

Reuse a pre-trained LLM; bridge text and signal.

## 3 · VFM

Render the series as an image; reuse a vision model.

# Building a TSFM means solving four design challenges




## Corpus

Integrate + synthesize data

## Encoding

Turn a signal into tokens

## Architecture

Pick a Transformer backbone

## Tasks

Serve many tasks at once

CHALLENGE 2 · ENCODING

# How do you turn a continuous signal into tokens?

## Point-wise

Each timestamp is one token.

USED BY **Time-MoE**

## Patch-wise

Slice into short fixed windows; each window is a token. Most popular choice.

USED BY **TimesFM · Moirai · MOMENT**

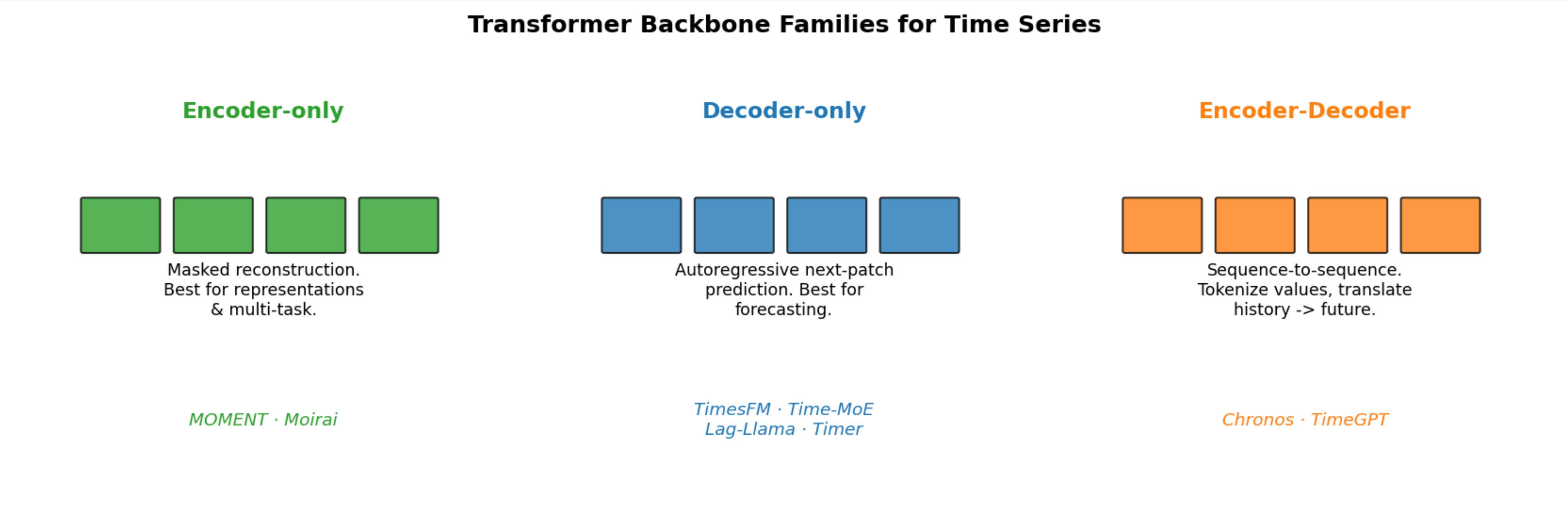
## Quantization-wise

Scale values and bin them into a fixed vocabulary — like words.

USED BY **Chronos**

*A second hidden choice: channel independence vs channel mixing — Moirai's "any-variate attention" handles any number of channels.*

# Three Transformer backbone families

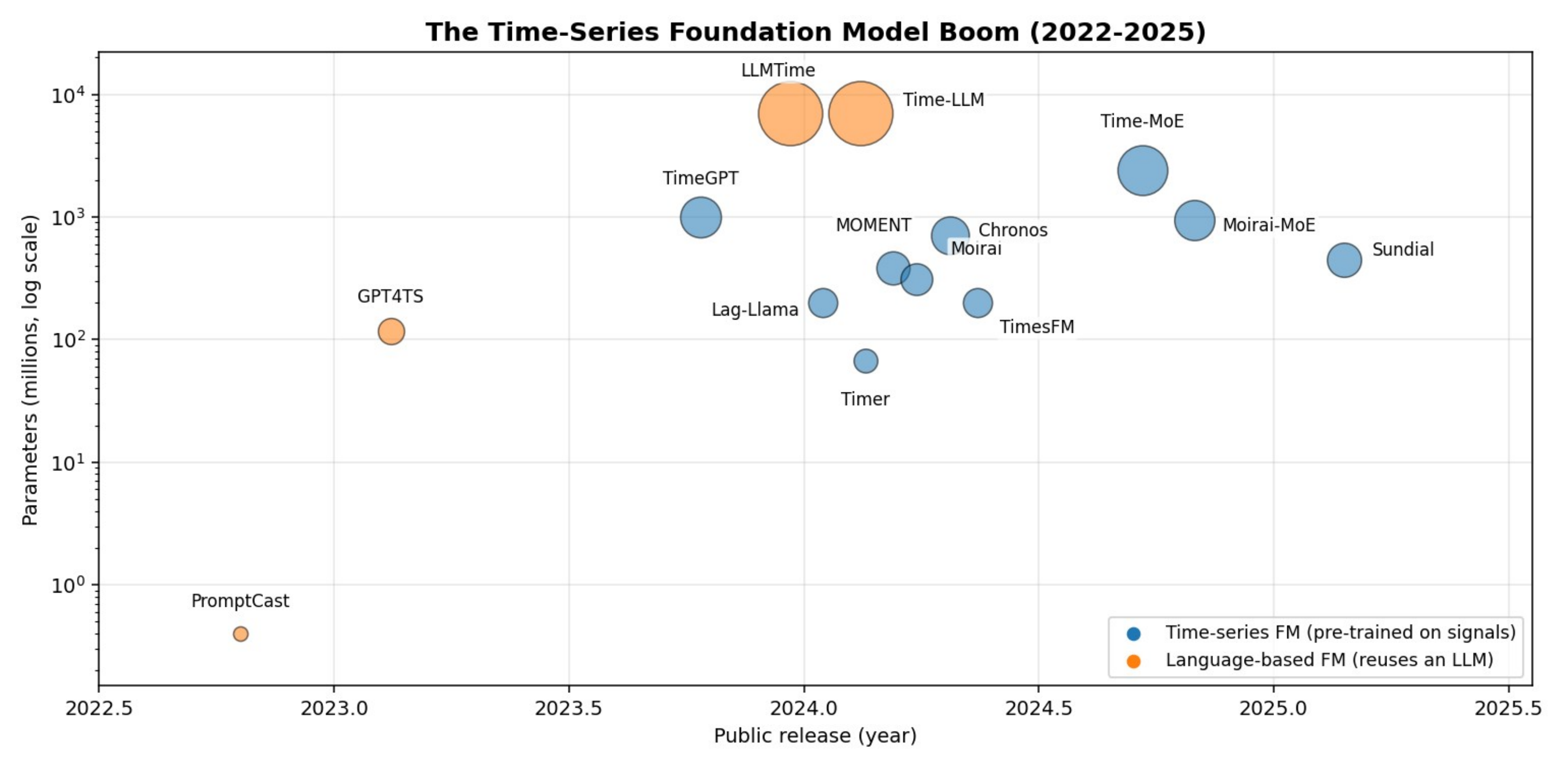



The pattern across all three surveys: **decoder-only for forecasting, encoder-only for representation & multi-task work.**

It is no accident that the largest TSFMs — Time-MoE, TimesFM — are decoder-only. Chronos is encoder-decoder because it reuses Google's T5 text model almost unchanged.

LANDSCAPE

# The model boom: ~100 models in two years




## What stands out

### Scale jump

67M → 2.4B parameters in under a year.

### Vendor land-grab

Google, Amazon, Salesforce, Nixtla, IBM, Tsinghua.

### Bigger ≠ better

IBM's Tiny Time Mixers (~1M params) competes — lightweight design is a real research track.

REFERENCE

# The headline TSFMs at a glance

| Model | Builder | Architecture | Tokenization | Scale | Pre-training corpus |
|---|---|---|---|---|---|
| **TimesFM** | Google | Decoder-only | Patch-wise | 200M | ~100B points |
| **Chronos** | Amazon | Encoder-decoder (T5) | Quantization-wise | 710M | ~84B points |
| **Moirai** | Salesforce | Encoder-only | Patch-wise | 311M | LOTSA, 27B points |
| **MOMENT** | CMU Auton Lab | Encoder-only | Patch-wise | 385M | Time Series Pile, 1.2B |
| **Time-MoE** | Time-MoE team | Decoder-only + MoE | Point-wise | 2.4B | Time-300B, 300B points |
| **Lag-Llama** | ServiceNow et al. | Decoder-only | Point-wise + lags | 200M | ~352M points |
| **Timer** | Tsinghua THUML | Decoder-only | Patch-wise | 67M | UTSD, ~1B points |

*Decoder-only models dominate pure forecasting; encoder-only models dominate representation and multi-task use. Corpus sizes span six orders of magnitude.*

# Borrowing brains from language and vision

## Language-based models (LFMs)

Can a model trained only on text forecast numbers? Surprisingly often, yes. Organized by how they bridge signal and language:

- LLMTime — write numbers as digit strings, let GPT continue them.
- Time-LLM — keep the LLM frozen, “reprogram” the series via cross-attention to word embeddings.
- GPT4TS — freeze most of GPT-2, tune only embeddings & norms; still handles 4 tasks.

*Recurring theme: parameter-efficient tuning.*

## Vision-based models (VFMs)

The survey's most surprising chapter: forecasting becomes image completion.

- Render the series as a line plot, heatmap or scalogram.
- VisionTS feeds it to a masked autoencoder pre-trained only on natural images.
- The model “inpaints” the missing future — and it works.

*My take: underrated — it quietly reuses the largest pre-trained models on Earth.*

# Putting the zero-shot claim to the test

## THE EXPERIMENT

**Model:** Chronos (chronos-bolt-small, ~48M params). Never call .fit() — pure zero-shot.

**Data:** 4 classic public datasets the model never saw — airline passengers, car sales, sunspots, daily temperatures.

**Baselines:** Naive, Seasonal Naive, Exponential Smoothing — each fit individually to its own series.

**Metric:** MASE — below 1.0 beats a seasonal-naive forecast; comparable across datasets.

## The fair fight

Classical baselines get an unfair advantage — they are trained on the very series they predict. The foundation model gets nothing.

**Baselines: fit per series**

**Chronos: zero training, zero tuning**

*Runs on a laptop CPU in about one minute. Code + executed notebook in the repo's reproduction/ folder.*

MY REPRODUCTION · RESULT

# Zero-shot beat models tuned to each series

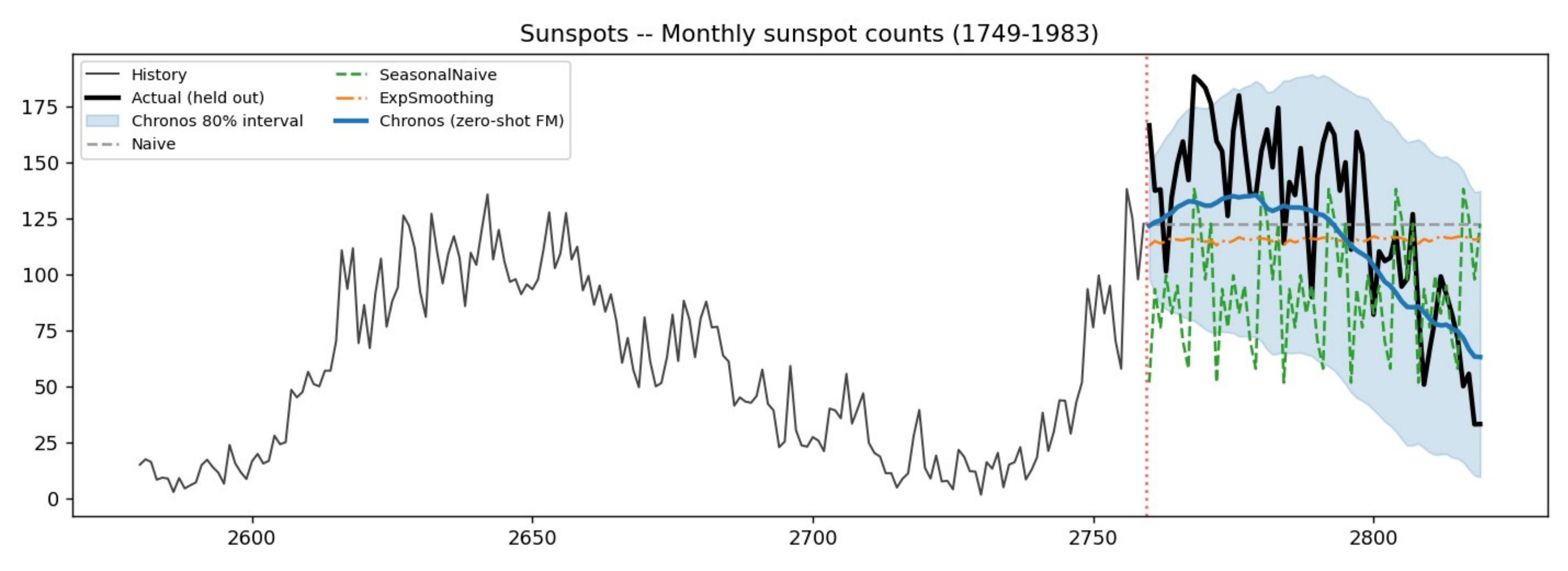
| ~1.66 | ~1.04 | ~0.92 |
|---|---|---|
| Seasonal Naive<br>mean MASE | Exp. Smoothing<br>mean MASE (fit per series) | Chronos zero-shot<br>mean MASE |
| 0 / 4 wins | 1 / 4 wins | 2 / 4 wins |

## The hypothesis holds.

A 48M-parameter model that did zero training on these series achieved a lower average error than Exponential Smoothing models each tuned to their own data — in about a second of CPU inference, with no training loop at all. It also produced calibrated uncertainty bands for free.

MY REPRODUCTION · EVIDENCE

# The clearest win: a long-horizon forecast




## Sunspots

60-step horizon

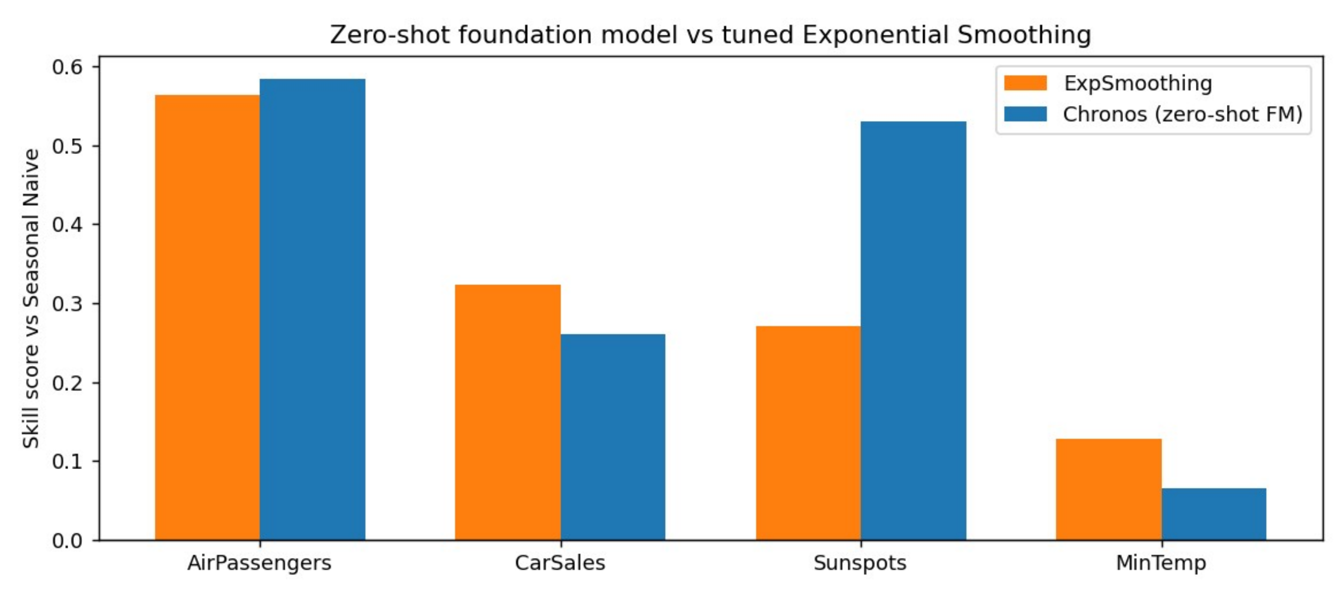
Chronos (blue) anticipates the falling solar cycle.

Classical baselines collapse to a flat line.

The shaded band is its 80% prediction interval — uncertainty for free.

*MASE 1.01 — the single best score in the whole experiment.*

# Strong — but not magic



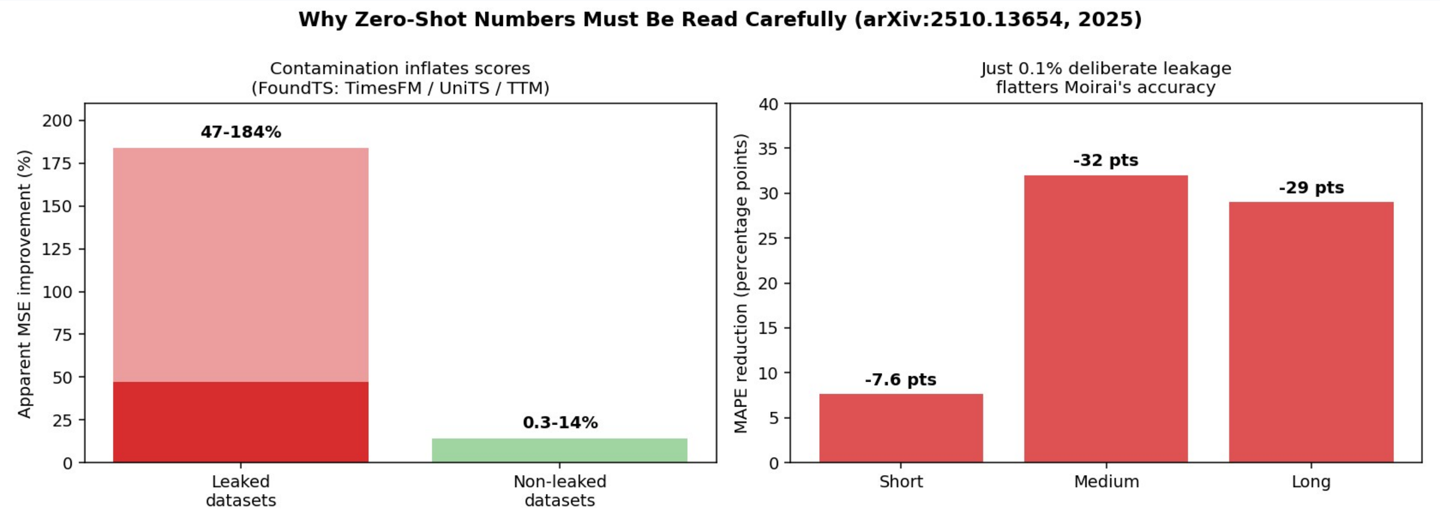

## Where it wins

Long horizons and rich structure — sunspots, airline passengers. Broad pre-training pays off exactly where classical methods struggle.

## Where it doesn't

On the near-stationary daily-temperature series, the plain Naive forecast won. The survey's “no free lunch” caveat in miniature — an honest review must say so.

# Can we even trust the benchmarks?




## Test-set contamination

If a model pre-trains on "every public dataset," the benchmark may already be in its training data.

**Only 7% of datasets were never used for pre-training or fine-tuning by some model.**

0.1% deliberate leakage flattered Moirai by up to 32 MAPE points — and bigger models memorized more.

### THE FIX

*A continuously advancing temporal split — the test set is always genuine, unseen future.*

MY TWO CENTS

# What I actually think after reading three surveys

### 1 The zero-shot result is real

I went in skeptical — “fit per series” should win. It didn't. A TSFM is now a strong default baseline you call in one line of code.

### 2 Headline numbers are softer than they look

After the contamination data, I don't fully trust any leaderboard MASE I haven't checked myself. Accuracy is solved enough; evaluation is not.

### 3 The vision pathway is underrated

“Render as an image, inpaint the future” sounds like a hack — but it reuses the biggest pre-trained models on Earth.

### 4 The frontier is trust, not scale

The next milestone isn't a 10B-parameter model. It's one you can audit: clean benchmarks, calibrated uncertainty, interpretability.

CONCLUSION

# Pre-train once, predict anything — finally, for time series too.

- **The map** — Three pathways, four design challenges, three architecture families — ~100 models in two years.

- **The proof** — My reproduction confirmed the core claim: a zero-shot foundation model matched models tuned per series.

- **The open problem** — Trustworthy evaluation. Until benchmarks are contamination-proof, every impressive number deserves a second look.

*The work that remains is making sure we can believe the predictions.*

# References & Links


1. Ye, Yu, Zhang, Wang, Li, Tsung. “Empowering Time Series Analysis with Foundation Models: A Comprehensive Survey.” arXiv:2405.02358v3, 2025. (primary paper reviewed)
2. Kottapalli et al. “Foundation Models for Time Series: A Survey.” arXiv:2504.04011, 2025.
3. Meyer, Kaltenpoth, Zalipski, Müller. “Rethinking Evaluation in the Era of Time Series Foundation Models: (Un)Known Information Leakage Challenges.” arXiv:2510.13654, 2025.
4. Ansari et al. “Chronos: Learning the Language of Time Series.” TMLR, 2024.
5. Das, Kong, Sen, Zhou. “A Decoder-Only Foundation Model for Time-Series Forecasting” (TimesFM). ICML, 2024.


**GitHub repository — code, executed notebook, article, slides & video**

`github.com/NMemane1/CMPE258-Short-Story-TS-FM-DeepLearning`

Medium article + reproduction figures linked in the repository README.

**Thank you — Nikita Memane · CMPE 258, San José State University**